VON NEUMANN-MORGENSTERN (vNM) EXPECTED UTILITY THEORY

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Core Idea of von Neumann–Morgenstern (vNM) Expected Utility Theory

Basic Information

Founders: John von Neumann, Oskar Morgenstern

Original work: Theory of Games and Economic Behavior, published in 1944; supplementary appendix on expected utility added in 1947.

Nature of theory: A normative theory for choice under risk. It defines how fully rational decision-makers SHOULD choose, rather than perfectly describing real human behaviour.

Core logic

Decisions are not based directly on monetary payoffs. Every outcome is assigned a subjective utility value U.

The value of a risky prospect equals its expected utility:

EU = SUM [p_i × U(C_i)]

Decision-makers select the option with higher expected utility.

Standard method to measure vNM utility via indifference lotteries:

Define the best outcome and worst outcome and standardize utility:

U(best outcome) = 1

U(worst outcome) = 0

Find the critical probability where the decision-maker is indifferent between receiving a fixed certain outcome and a lottery.

This critical probability equals the utility of the certain outcome.

Numerical Demonstration

Three possible outcomes:

C1: Receive

$10,000 (best outcome), U(10000) = 1

C3: Receive $0 (worst outcome), U(0) = 0

C2: Receive $4,000 for sure (intermediate outcome)

Two alternatives:

A: Receive

$4,000 with certainty

B: Lottery: probability u of winning $10,000, probability (1-u) of winning $0

Suppose the critical point: decision-maker is indifferent between A and B when u = 0.6.

By definition: U(4000) = 0.6

Expected monetary value of lottery B: 0.6 × 10000 + 0.4 × 0 =

$6,000.

The agent accepts a certain $4,000 instead of a gamble with higher average money payoff. This behaviour represents risk aversion.

Decision-making using Expected Utility

Plan X: 60% chance of $10,000; 40% chance of $0

EU_X = 0.6 × 1 + 0.4 × 0 = 0.6

Plan Y: Certain $4,000

EU_Y = U(4000) = 0.6

EU values are equal; the decision-maker is indifferent.

Modified comparison:

Plan X: 50% chance of $10,000, 50% chance of $0 → EU_X = 0.5

Plan Y: Certain $4,000 → EU_Y = 0.6

Since EU_Y > EU_X, a risk-averse agent chooses the guaranteed payoff.

Risk preferences differ between individuals. A risk-seeking person might treat certain $4,000 as equivalent to a 50% chance of $10,000, meaning U(4000) = 0.5.

Decision rule is identical for everyone: maximize expected utility. Heterogeneity comes from different shapes of individual utility functions.

Expected Monetary Value (EMV) vs Expected Utility (EU)

Expected Monetary Value (EMV): Calculate only the mathematical average of money payoffs

EMV = SUM [p_i × Monetary Payoff_i]

Hidden assumption: every extra unit of money creates identical subjective satisfaction; utility function is linear: U(x) = x.

Expected Utility (EU): Compute weighted average of subjective satisfaction generated by wealth

EU = SUM [p_i × U(Payoff_i)]

Allows non-linear utility functions and can capture different risk attitudes.

Contradiction Example

Choice 1: Receive

$4,000 for sure

Choice 2: Lottery: 50% chance of $10,000, 50% chance of $0

EMV(lottery) = 0.5 × 10000 + 0.5 × 0 =

$5,000

Pure EMV rule recommends selecting the lottery. In reality, most ordinary people select the safe $4,000.

Conclusion: EMV cannot explain observed behaviour; expected utility theory uses diminishing marginal utility to resolve this conflict.

Three Types of Risk Attitudes (Distinguished by Utility Function Shape)

Risk aversion (typical for most people)

U(E[X]) > EU(X); concave utility function with diminishing marginal utility.

The utility of receiving the expected value for sure is higher than expected utility of the gamble. People accept a fixed

income lower than EMV to avoid risk.

Risk neutrality

U(x) = x, linear utility. EU equals EMV. Only concerned with average monetary outcome. Widely used for modelling large corporations.

Risk seeking (gambler behaviour)

U(E[X]) < EU(X); convex utility function. People are willing to accept gambles with negative expected monetary value.

Marginal Utility

Definition: Extra utility created by one additional unit of wealth

MU = $\Delta U / \Delta X$

Diminishing Marginal Utility: As total wealth rises, the satisfaction brought by each additional dollar gradually falls.

Example:

The first $1,000 earned when a person has zero wealth delivers very high utility. An extra $1,000 when the person already owns $9,000 creates much smaller additional happiness.

Historical origin:

Daniel Bernoulli applied diminishing utility of wealth to solve the St. Petersburg Paradox in 1738.

Marginal Revolution (1870s): Jevons, Menger, Walras established modern marginal utility theory.

Inference: The pain caused by a monetary loss is larger than the pleasure created by an equal-sized monetary gain.

St. Petersburg Paradox

Proposed by Nicolaus Bernoulli (1713). Daniel Bernoulli provided a solution in 1738.

Game rules: Toss a fair coin repeatedly.

If the first head appears on the n-th trial, prize = 2^n dollars; corresponding probability = 1 / (2^n).

Expected monetary value:

EMV = SUM from n=1 to infinity of [(1 / 2^n) × 2^n] = 1 + 1 + 1 + ... → infinity.

EMV logic implies people should pay any finite amount to enter this game.

Paradox: Almost no individual is willing to pay more than roughly $20 to participate.

Bernoulli's solution: People decide based on utility rather than raw money. Wealth has diminishing marginal utility. He adopted logarithmic utility U(x) = ln(x), proving expected utility is finite.

Important distinction: Bernoulli relied on intuitive utility without formal axioms. The full axiomatic foundation for expected utility was built over two centuries later by von Neumann and Morgenstern.

Four vNM Axioms of Expected Utility

If preferences over lotteries satisfy all four axioms, the decision-maker behaves as if maximizing expected utility.

Notation: $\succ$ = strictly preferred; ~ = indifferent.

Completeness

For any two lotteries A and B: either A $\succ$ B, or B $\succ$ A, or A ~ B.

Meaning: The decision-maker can rank every pair of risky alternatives.

Transitivity

If A $\succ$ B and B $\succ$ C, then A $\succ$ C.

Violating transitivity creates cyclic preferences and generates money pump arbitrage.

Independence Axiom (most frequently challenged)

If A $\succ$ B, for any probability 0 < p < 1 and any lottery C:

pA + (1-p)C $\succ$ pB + (1-p)C

Plain interpretation: Adding the same random component C to both options does not reverse the original preference ranking.

The Allais Paradox directly contradicts the independence axiom.

Continuity

If A ≻ B ≻ C, there must exist some probability p such that:

pA + (1-p)C ~ B

This axiom supports the indifference lottery method to measure vNM utility values.

Classic Application: Using Expected Utility to Explain Insurance Demand

Baseline setup: Initial wealth =

$100,000

Risk: 10% probability of fire, wealth falls to $40,000; 90% probability of no fire, wealth unchanged.

Insurance premium = $6,500. Buying insurance completely eliminates fire risk, final fixed wealth = $93,500.

Step 1: Calculate EMV

Without insurance:

EMV = 0.9 × 100000 + 0.1 × 40000 =

$94,000

With insurance: fixed wealth equals $93,500.

The uninsured option has higher expected money value.

Pure EMV theory predicts zero demand for insurance, which conflicts with real-world observation.

Step 2: Expected Utility Analysis for Risk-Averse Agents

Standardization: U(100000) = 1, U(40000) = 0

Assume calibrated preference: possessing $93,500 for certain is indifferent to a lottery with 92% chance of $100,000 and 8% chance of $40,000. Therefore U(93500) = 0.92.

EU(no insurance) = 0.9 × 1 + 0.1 × 0 = 0.9

EU(buy insurance) = 0.92

EU with insurance is higher; risk-averse agents choose to purchase cover.

Core reasoning: The unhappiness from large wealth losses outweighs the joy of saving insurance premiums. Diminishing marginal utility makes people willing to pay risk premiums to avoid catastrophic outcomes.

Boundary condition: People will not buy insurance at arbitrarily high premiums. If premiums become too expensive or disaster probability extremely low, expected utility of self-insurance becomes higher.

Two Famous Paradoxes Challenging vNM Theory

7.1 Allais Paradox (Violates Independence Axiom)

First set of choices

A: 100% chance to receive

$1 million

B: 10% chance of $5 million, 89% chance of $1 million, 1% chance of $0

Most experimental subjects choose A.

Second set of choices

C: 11% chance of $1 million, 89% chance of $0

D: 10% chance of $5 million, 90% chance of $0

Most experimental subjects choose D.

This combination of choices violates the vNM independence axiom. Stable observed choice patterns deviate from theoretical predictions.

7.2 Ellsberg Paradox

Concept distinction:

Risk: Probabilities are objectively known (fair coin flip)

Ambiguity: Probabilities are unknown.

Experiment setup: An urn contains 90 balls. 30 are red. The remaining 60 balls are black and yellow in unknown proportions.

Choice group one: Bet on red to win $100; bet on black to win $100. Most participants prefer red.

Choice group two: Bet on red or yellow; bet on black or yellow. Most participants prefer black or yellow.

These preference rankings cannot be explained by standard expected utility with fixed probabilities. The pattern is known as ambiguity aversion.

Limitation of basic vNM theory: The framework only handles risk with known probabilities. It cannot model ambiguous situations where odds are unknown.

Extensions and Competing Decision Theories

vNM expected utility acts as the standard benchmark of rational choice (normative theory). Laboratory experiments repeatedly show systematic violations of its axioms. Researchers developed alternative descriptive theories to match real human behaviour:

Prospect Theory (Kahneman & Tversky)

Agents evaluate gains and losses relative to a reference point, not final total wealth

Loss aversion: the pain of losses exceeds satisfaction from equivalent gains

Probability distortion: people overweigh tiny probabilities and underweight moderate probabilities

Explains Allais Paradox, insurance purchase and lottery participation.

Rank-Dependent Utility (RDU)

Preserves the utility function structure but adds a probability weighting function. People subjectively adjust weights for extreme outcomes. This modification addresses problems with the independence axiom.

Transitivity Axiom Foundation: Money Pump Thought Experiment

Definition

The money pump is a core thought experiment in decision theory.

If an agent holds cyclic intransitive strict preferences:

$A > B, B > C, C > A$

A trader can construct sequential trades to repeatedly extract wealth from the decision-maker.

Simple Example

Preferences: Apple > Banana, Banana > Orange, Orange > Apple. Initial endowment: Orange.

Pay

$1 to exchange Orange for Apple

Pay another $1 to exchange Apple for Banana

Pay another $1 to exchange Banana for Orange

After one full cycle: the agent returns to holding the original Orange, with a net loss of $3. The exploitation can continue indefinitely if preferences stay unchanged.

Core Inference for the transitivity axiom

Agents without transitive preferences are vulnerable to money pump exploitation.

Important Qualifications

Strict cyclic preferences create money pumps; indifference cycles do not.

This is an idealized theoretical case. In reality, agents learn to refuse repeated trades and transaction costs limit endless arbitrage, so permanent infinite money pumps rarely exist.

Many laboratory experiments document stable cyclic preferences over risky lotteries. These findings are evidence against the full rationality assumption in neoclassical economics.

Classification

Weak money pump: wealth falls after a single cycle

Strong money pump: exploitation can repeat infinitely (classic version)

Supplementary: Empirical Cyclic Lottery Preferences in Laboratory Studies

Many controlled experiments observe stable intransitive rankings over risky lotteries:

Tversky (1969) landmark experiment

Subjects evaluate pairs of lotteries. When lotteries are similar, people focus on winning probability. When lotteries differ widely, people focus on prize size. This generates consistent cycles such as P2 > P3, P3 > P4, P4 > P2.

Classroom three-lottery demonstration

X: High probability, small prize

Y: Low probability, large prize

Z: Medium probability, medium prize

Common observed cycle: X > Y, Y > Z, Z > X.

Academic debate:

Neoclassical viewpoint: Cyclic preferences represent irrationality and expose agents to money pumping.

Behavioural economics viewpoint: Human preferences are context-dependent and change with framing. Persistent fixed cycles are difficult to sustain in real markets, limiting practical money pump risks.